

Guardian Engine Engineering Handbook

Version 0.1

Welcome

Welcome to Guardian Engine.

We are not building an application.

We are building a platform that may operate for decades.

Every engineering decision should improve the platform.

Never optimize for today at the expense of tomorrow.

Engineering Philosophy

We solve problems.

We do not build features.

We automate workflows.

We reduce cognitive load.

Every line of code must have a business reason.

Golden Rule

If a feature cannot explain:

«"What user problem does this solve?"»

It should not exist.

Product Hierarchy

Guardian Engine

↓

Products

↓

Modules

↓

Workflows

↓

Components

↓

Functions

Never design in reverse order.

Technology Principles

Priorities:

1. Reliability
2. Maintainability
3. Security
4. Simplicity
5. Performance

Premature optimization is forbidden.

Architecture

Layers

Presentation

↓

Application

↓

Domain

↓

Infrastructure

Dependencies always point inward.

React must never contain business logic.

Domain First

Before creating a component:

Create the domain model.

Before creating the API:

Create the workflow.

Before creating the database:

Create the entities.

Workflow Driven Development

Wrong

InspectionScreen.tsx

Correct

InspectionWorkflow

Everything begins with a workflow.

AI Rules

AI never becomes the source of truth.

AI always works on verified context.

If verified knowledge exists:

Return verified knowledge.

Only then:

Use AI for explanation.

Never invent legal information.

Knowledge Rules

Every legal entry must contain:

Country

Language

Source

Verification Date

Version

Confidence

Effective Date

Last Review

Knowledge without metadata does not exist.

Decision Engine Rules

Decision Engine is deterministic.

No LLM.

No probabilistic answers.

Only rules.

Example:

Country

↓

Workflow

↓

Situation

↓

Result

Coding Rules

Small files.

Small classes.

Pure functions whenever possible.

Dependency Injection.

Interface First.

Avoid Singleton.

Avoid God Objects.

Prefer composition.

Naming

Good

KnowledgeRepository

InspectionWorkflow

DecisionEngine

CountryResolver

EmergencyCard

Bad

Utils

Helper

Common

Manager

Stuff

DataService

Folder Structure

apps/

core/

packages/

shared/

docs/

tests/

scripts/

No business logic inside UI folders.

Git Rules

One feature

=

One branch

One Pull Request

=

One purpose

Never mix refactoring and new features.

Pull Requests

Every PR must answer:

Why?

What changed?

Risk?

Rollback?

Future impact?

ADR

Every important decision must have an ADR.

Template:

Context

Decision

Alternatives

Consequences

Status

Security

Encrypt local data.

Never trust client input.

Never expose secrets.

Validate everything.

Log safely.

No personal data inside logs.

Offline

Offline is a feature.

Not an afterthought.

Emergency workflows must never require Internet.

User Experience

Under stress:

Maximum

2 taps

Maximum

1 decision

Maximum

5 seconds to reach critical information.

Error Handling

Never expose technical errors.

Always explain:

What happened.

What user should do.

How to recover.

Performance

Measure first.

Optimize later.

Readable code is preferred over clever code.

Testing

Business Rules

100%

Decision Engine

100%

Workflow Engine

100%

UI

Only critical paths.

Documentation

Every engine

must contain

README

Architecture

ADR

Examples

No undocumented module may enter production.

Engineering Culture

Question assumptions.

Prefer evidence.

Keep discussions respectful.

Disagree with ideas.

Never with people.

Final Principle

We are not shipping code.

We are building trust.

Every release must make the user feel safer than before.

If it does not increase trust,

it should not be released.